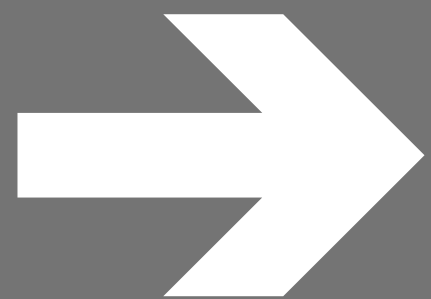


ML Project



**CLASSIFICATION &
REGRESSION**

DESERT NINJAS





Table of Content



Team Members



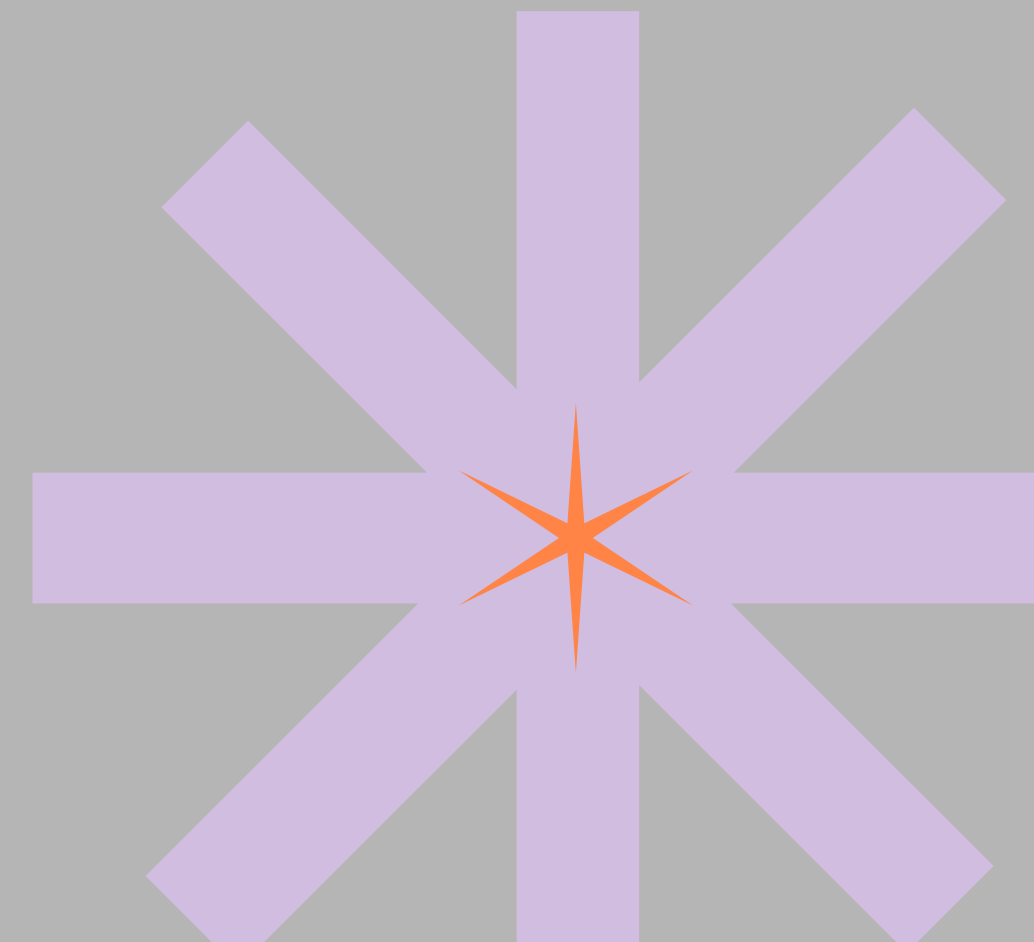
Dataset Overview



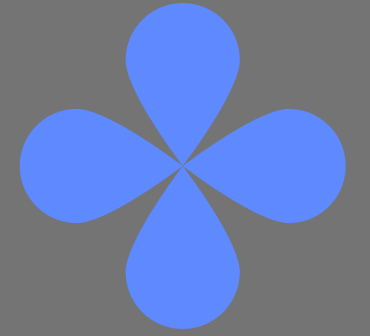
Feature Engineering



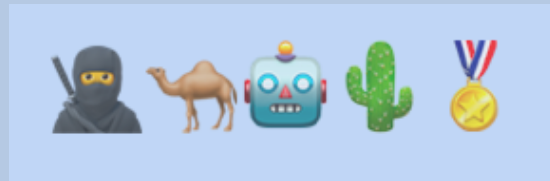
ML Models



Team Members

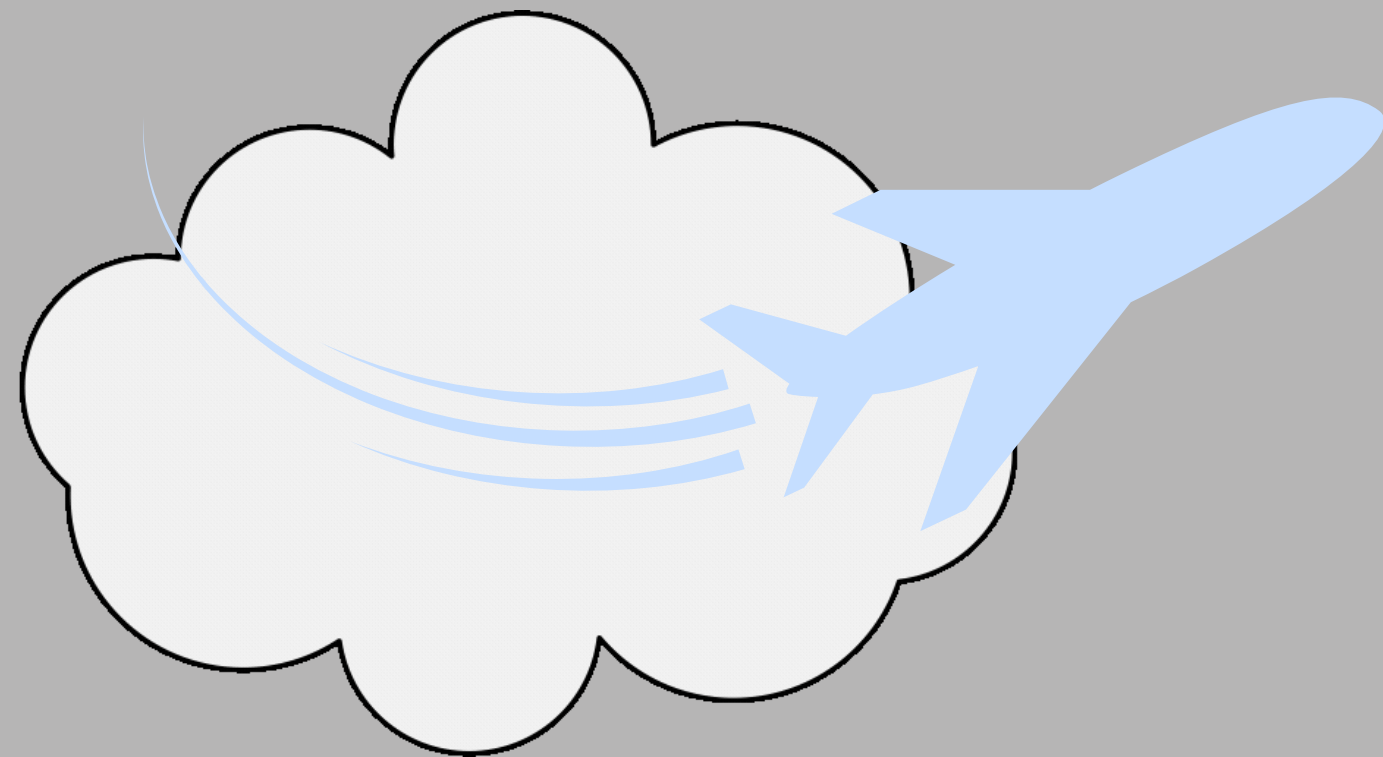


**DESERT
NINJAS**



Reema Alaswad
Maha Alhazzani
Aljohara Alkanhal
Raghad Aleisa
Eman Aldosari





PLEASE FASTEN YOUR SEATBELTS WE ARE
READY TO *TAKE-OFF*



Dataset Overview

THIS DATASET CONTAINS ALL FLIGHT INFORMATION INCLUDING **CANCELLATIONS**
AND **DELAYS** BY *AIRLINE* FOR DATES BACK TO JANUARY 2022.

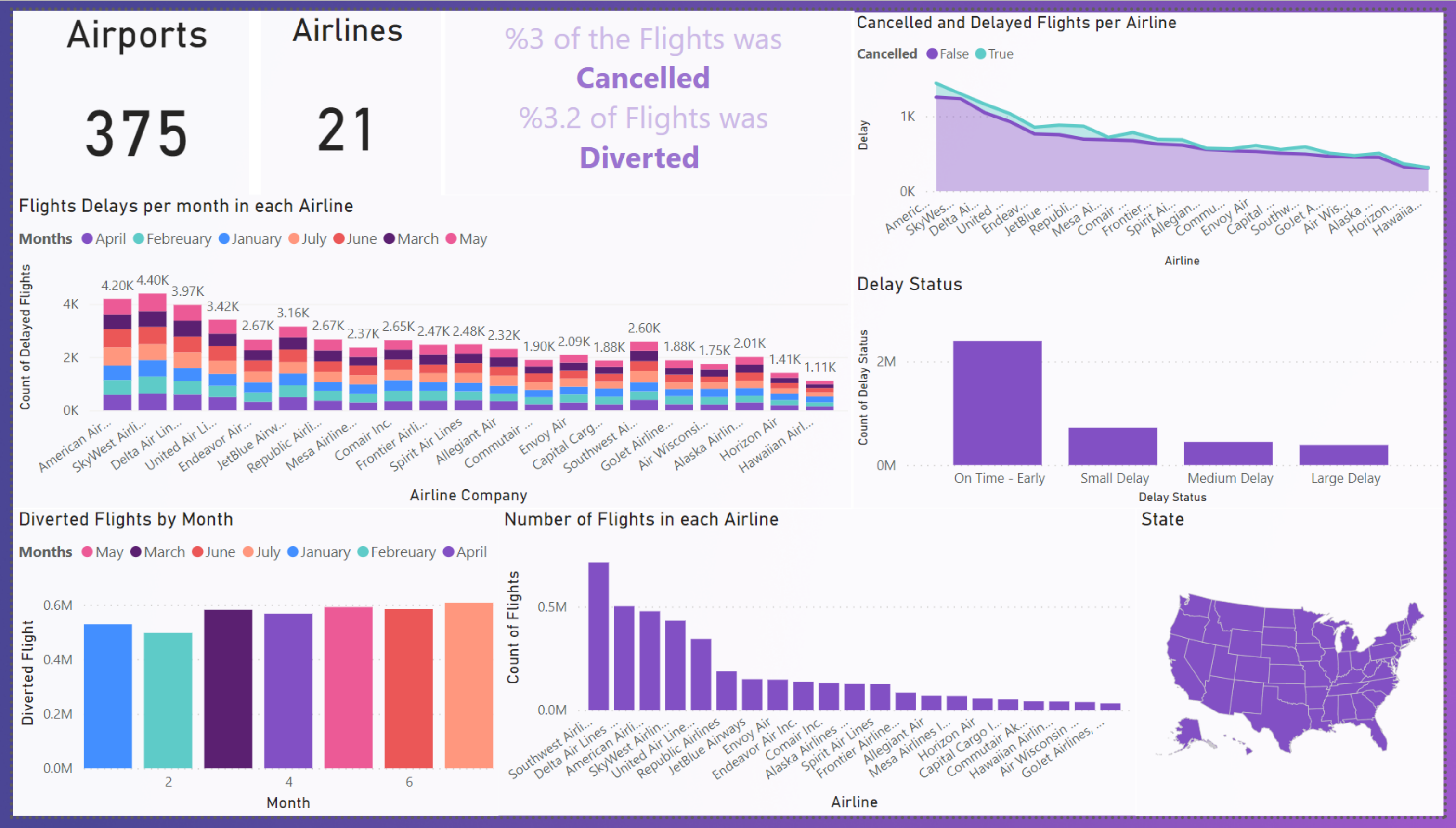
375 AIRPORTS

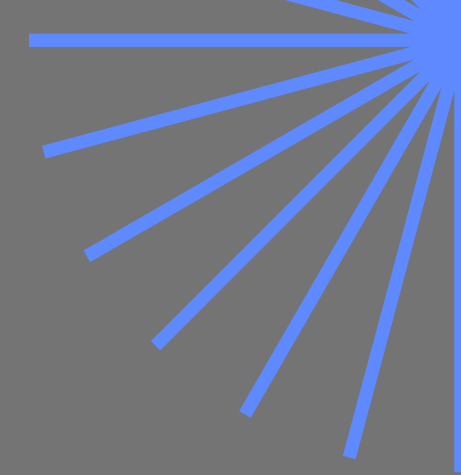
21 AIRLINES

+40M FLIGHTS

61 COLUMNS

Dashboard





ML Models

- ✦ Logistic Regression
 - ✦ Random Forest
 - ✦ Multi-linear Regression
- 

Feature Engineering



Feature Engineering Steps

**Delete
Unneeded
Columns**

**Handling
Missing
Values**

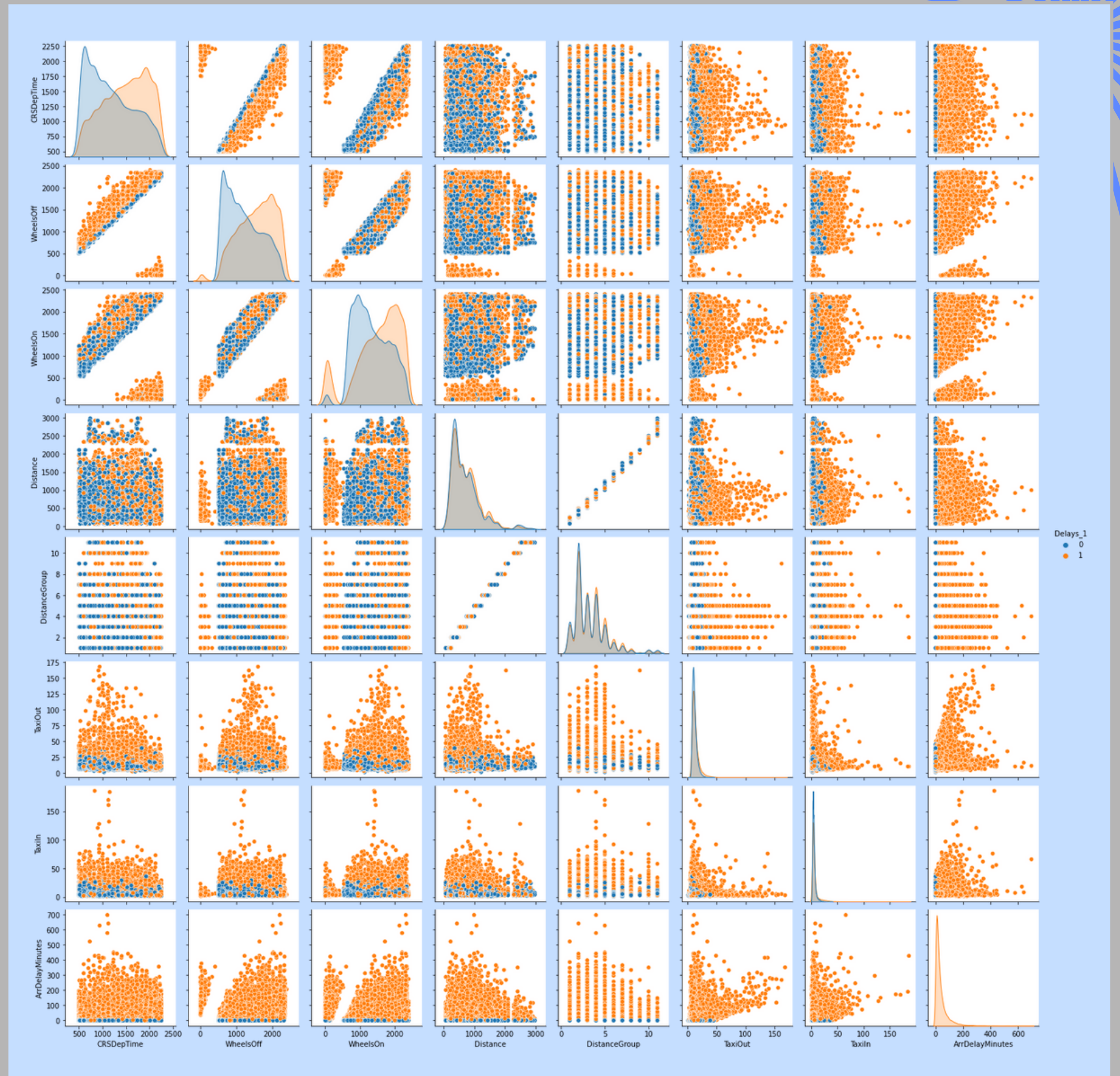
**Binning the
Continues
Target**

**Ordinal
Encoder**



Pairplot

```
sns.pairplot(df, hue = 'Delays_1' )
```



Correlation

```
corr = df.corr()
```

```
corr.style.background_gradient(cmap='coolwarm')
```

	CRSDepTime	WheelsOff	WheelsOn	Distance	DistanceGroup	TaxiOut	TaxiIn	ArrDelayMinutes	Delays_1
CRSDepTime	1.000000	0.918696	0.538781	-0.071714	-0.071795	-0.076998	-0.054743	0.112062	0.244285
WheelsOff	0.918696	1.000000	0.599007	-0.044683	-0.044301	-0.031727	-0.034176	0.185573	0.290672
WheelsOn	0.538781	0.599007	1.000000	0.007348	0.005489	0.014367	0.010860	0.044720	0.147258
Distance	-0.071714	-0.044683	0.007348	1.000000	0.988493	-0.018372	0.049108	0.093347	0.069288
DistanceGroup	-0.071795	-0.044301	0.005489	0.988493	1.000000	-0.019358	0.048789	0.092730	0.069743
TaxiOut	-0.076998	-0.031727	0.014367	-0.018372	-0.019358	1.000000	0.028227	0.194085	0.113853
TaxiIn	-0.054743	-0.034176	0.010860	0.049108	0.048789	0.028227	1.000000	0.188383	0.121288
ArrDelayMinutes	0.112062	0.185573	0.044720	0.093347	0.092730	0.194085	0.188383	1.000000	0.472906
Delays_1	0.244285	0.290672	0.147258	0.069288	0.069743	0.113853	0.121288	0.472906	1.000000

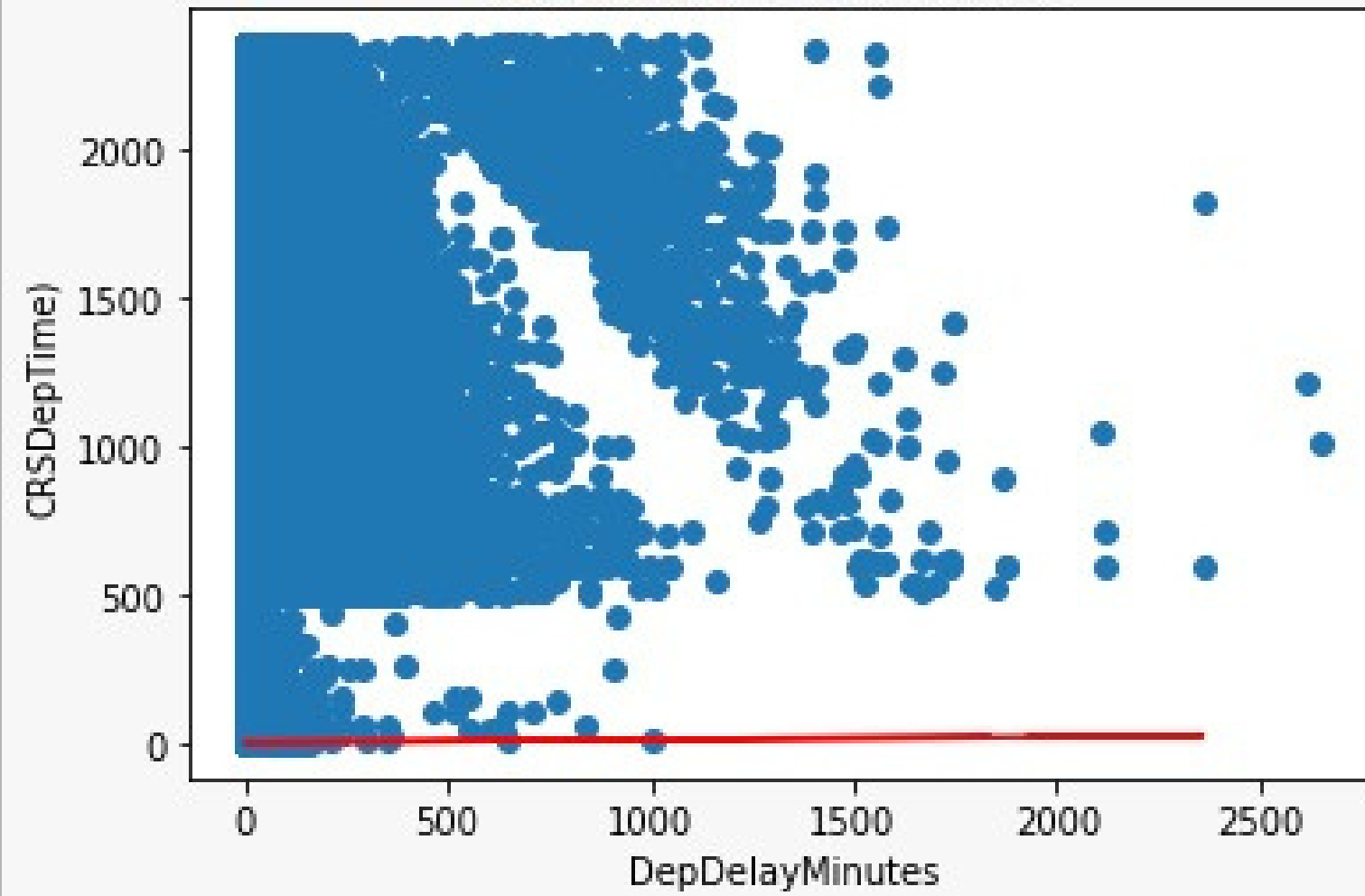
Challenges and Lessons Learned:



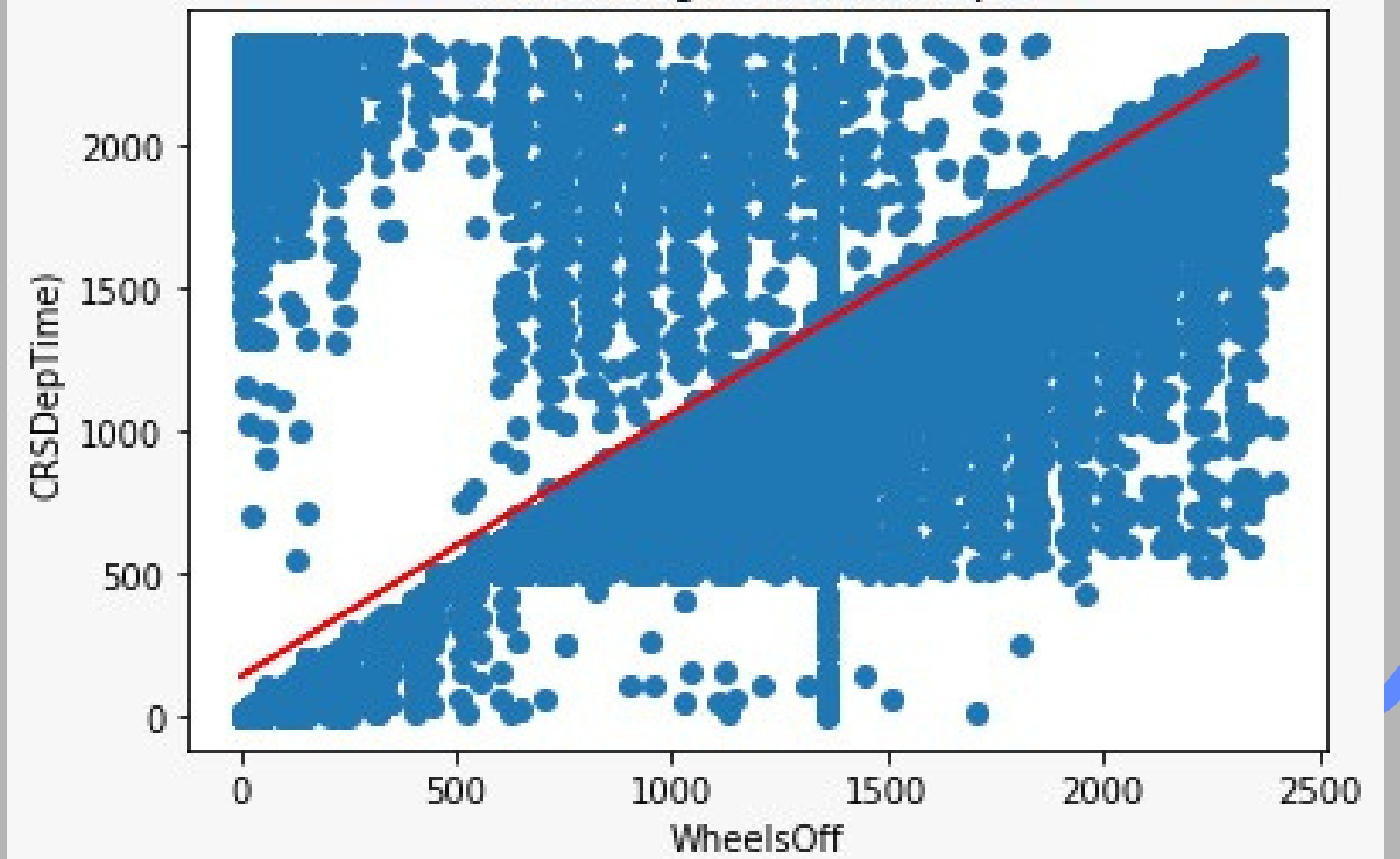
- ✦ Dataset Size (+4M rows, 61 columns)
- ✦ Choosing the Wrong Target (E.g., Cancelled/not cancelled)
- ✦ Data Leakage (Linear/Multi-Linear Regression Model)
- ✦ EDA Analysis (Time constrain)



Linear Regression Example



Linear Regression Example



IS THE FLIGHT GOING TO BE DELAYED?

BINARY CLASSIFICATION (LOGISTIC)



CONDITIONS:

No Delay/ Early arrival	0 mins	60%
Delayed	0+ mins of delay	40%
Target	Delays	
Features	"CRSDepTime", 'WheelsOff','WheelsOn','Distance','DistanceGroup', "TaxiOut", "TaxiIn"	

CONFUSION MATRIX

True label	119184	0
	3090	68987

Predicted label



IS THE FLIGHT GOING TO BE DELAYED?

RANDOM FOREST

CONDITIONS:

No Delay/ Early arrival	0 mins	60%
Delayed	0+ mins of delay	40%
Target	ArrDelayMinutes	
Features	"CRSDepTime", 'WheelsOff','WheelsOn','Distance','DistanceGroup', "TaxiOut", "TaxiIn"	



CONFUSION MATRIX

True label	119184	0
	0	72077
Predicted label		

IS THE FLIGHT GOING TO BE DELAYED?



LINEAR REGRESSION

Target

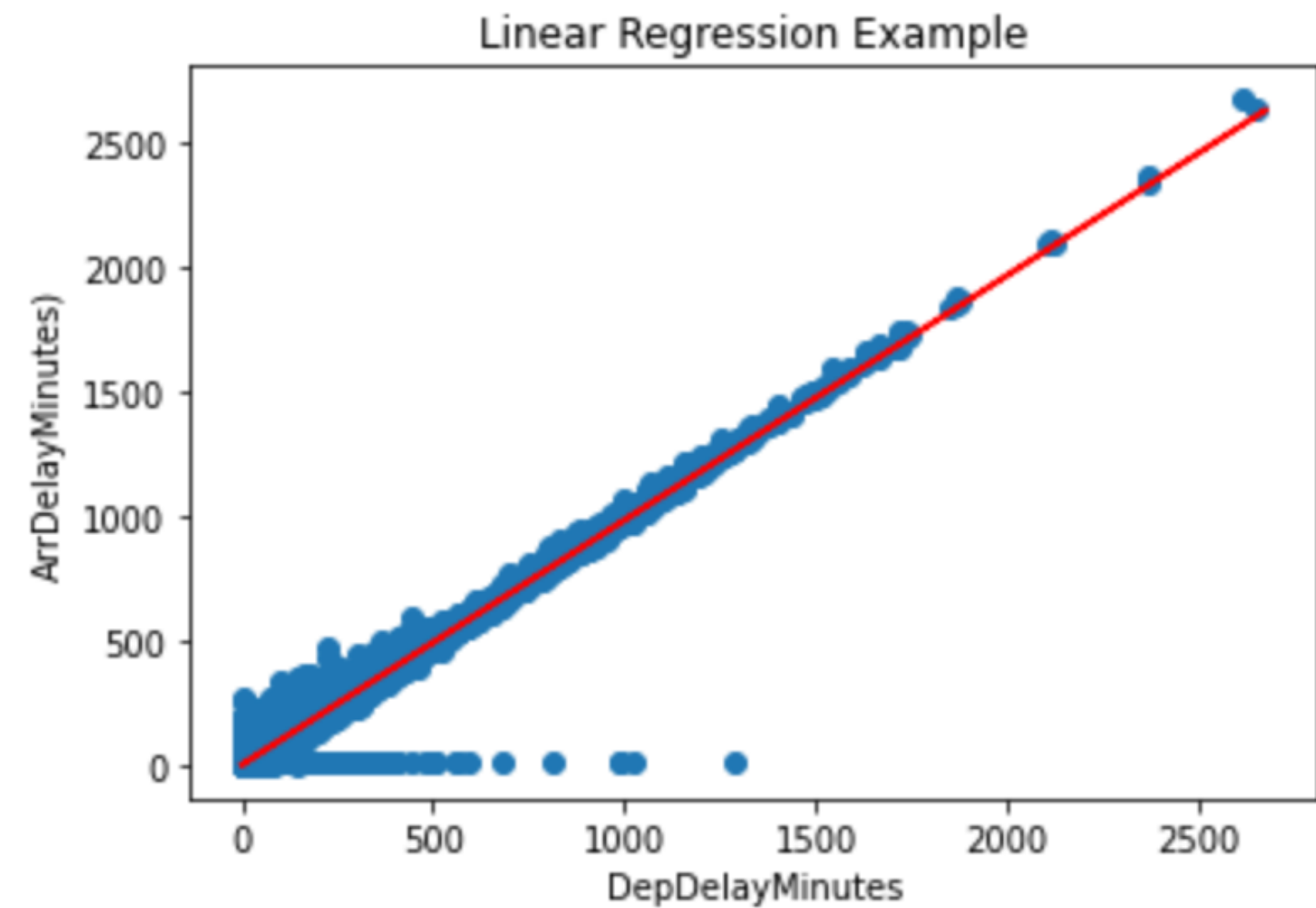
ArrDelayMinutes

Features

DepDelayMinutes

MEAN ABSOLUTE ERROR

5



IS THE FLIGHT GOING TO BE DELAYED?

MULTI-LEARNER REGRESSION

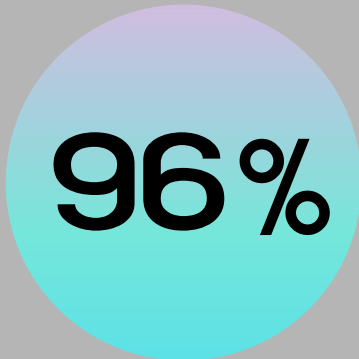


Target	ArrDelayMinutes
Features	"DepartureDelayGroups", "DepDel15", "DepDelay", "DepDelayMinutes"

MEAN ABSOLUTE ERROR



R2 SCORE



ألغي رحلتي



THANKS FOR LISTENING!

الحمد لله على سلامتنا!

SEE YOU ON ANOTHER TRIP!

